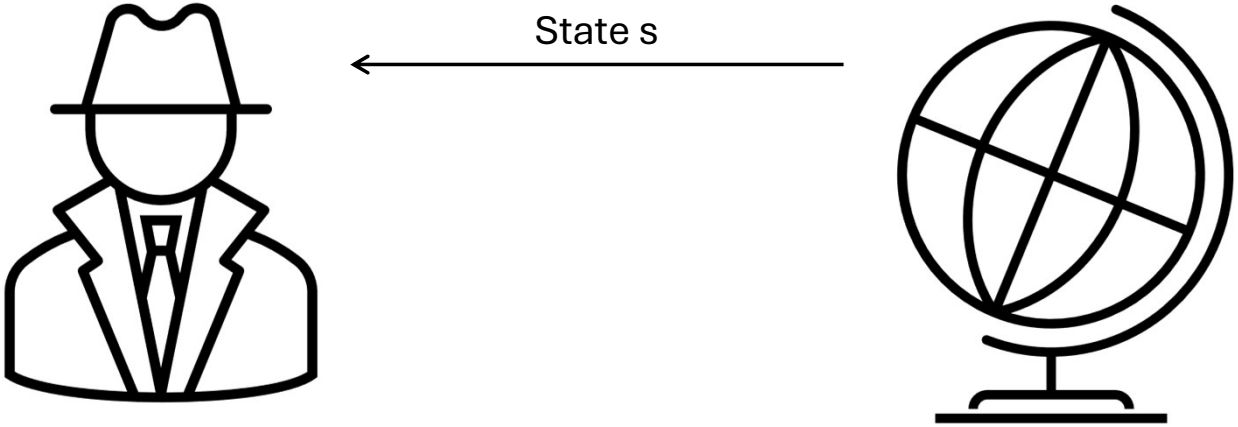
MARL
Incentive Design
Contract Theory

Jakub Tłuczek

Classic Reinforcement Learning



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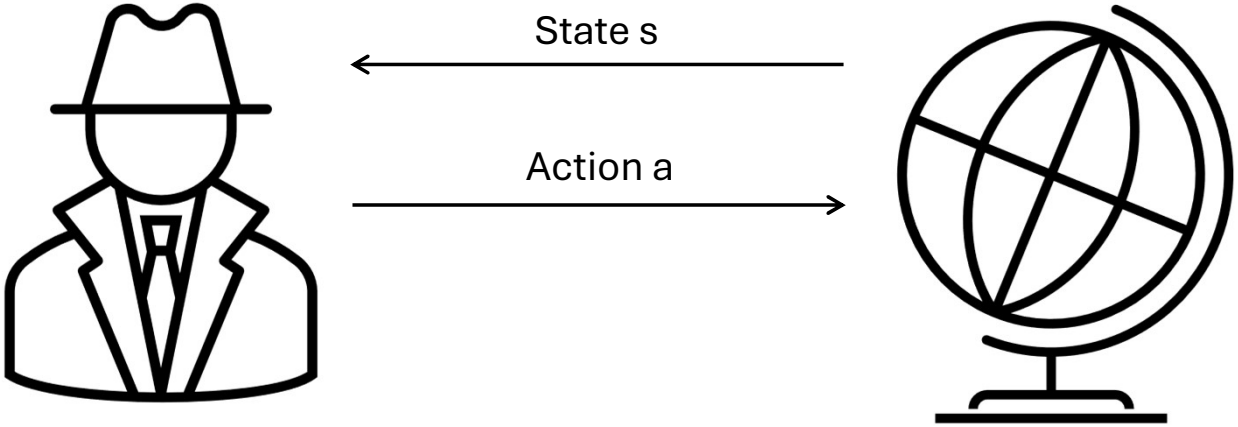

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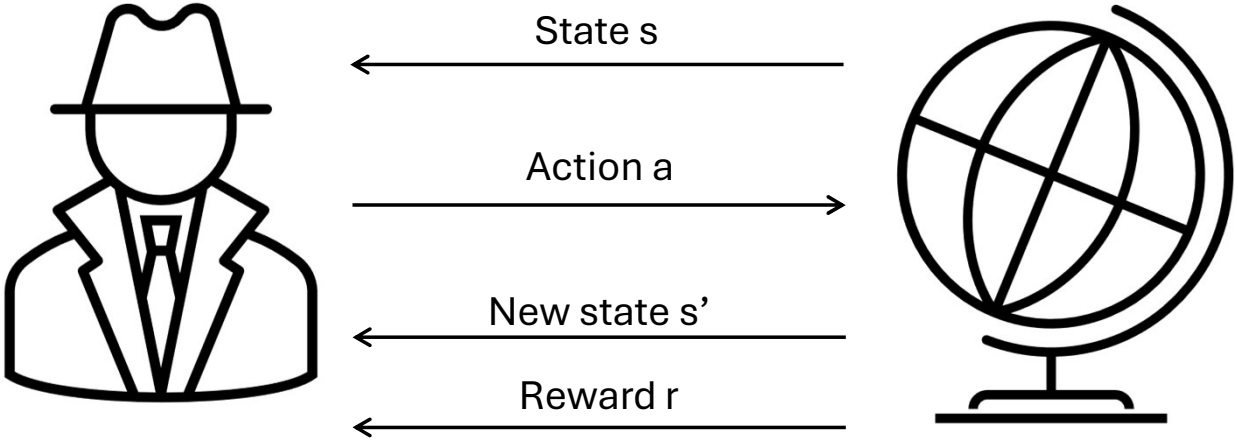
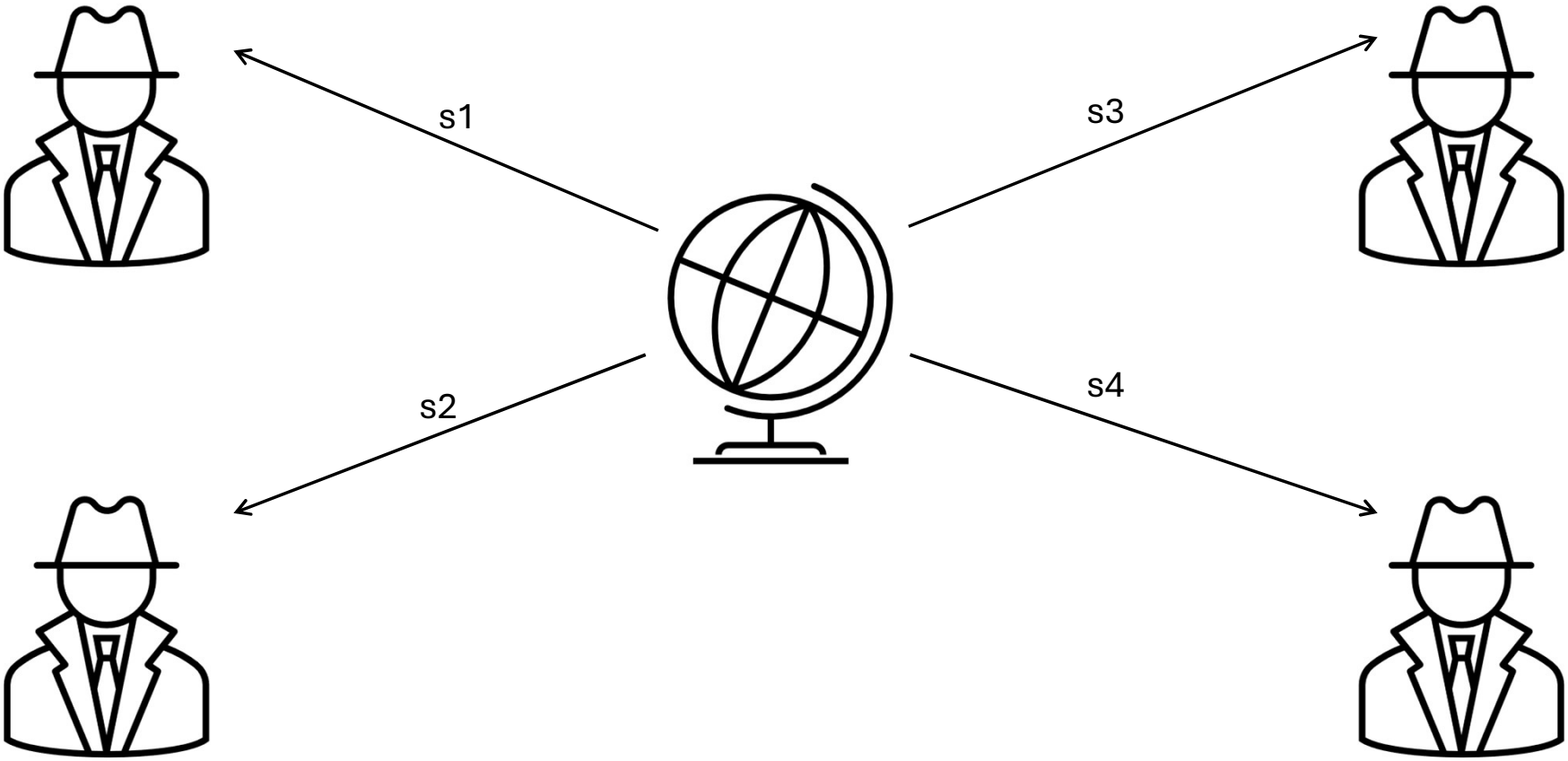


Reinforcement learning with multiple agents



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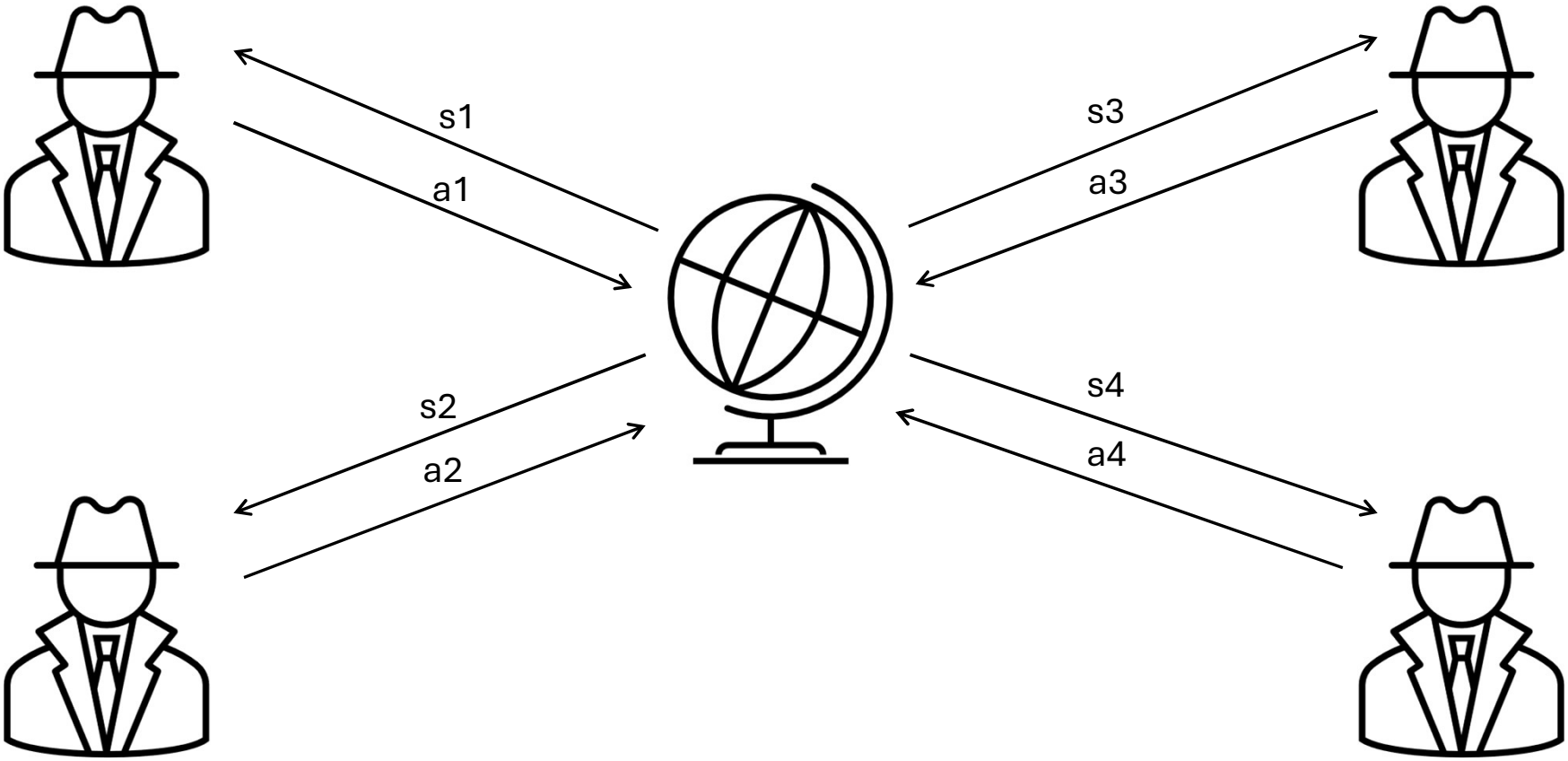

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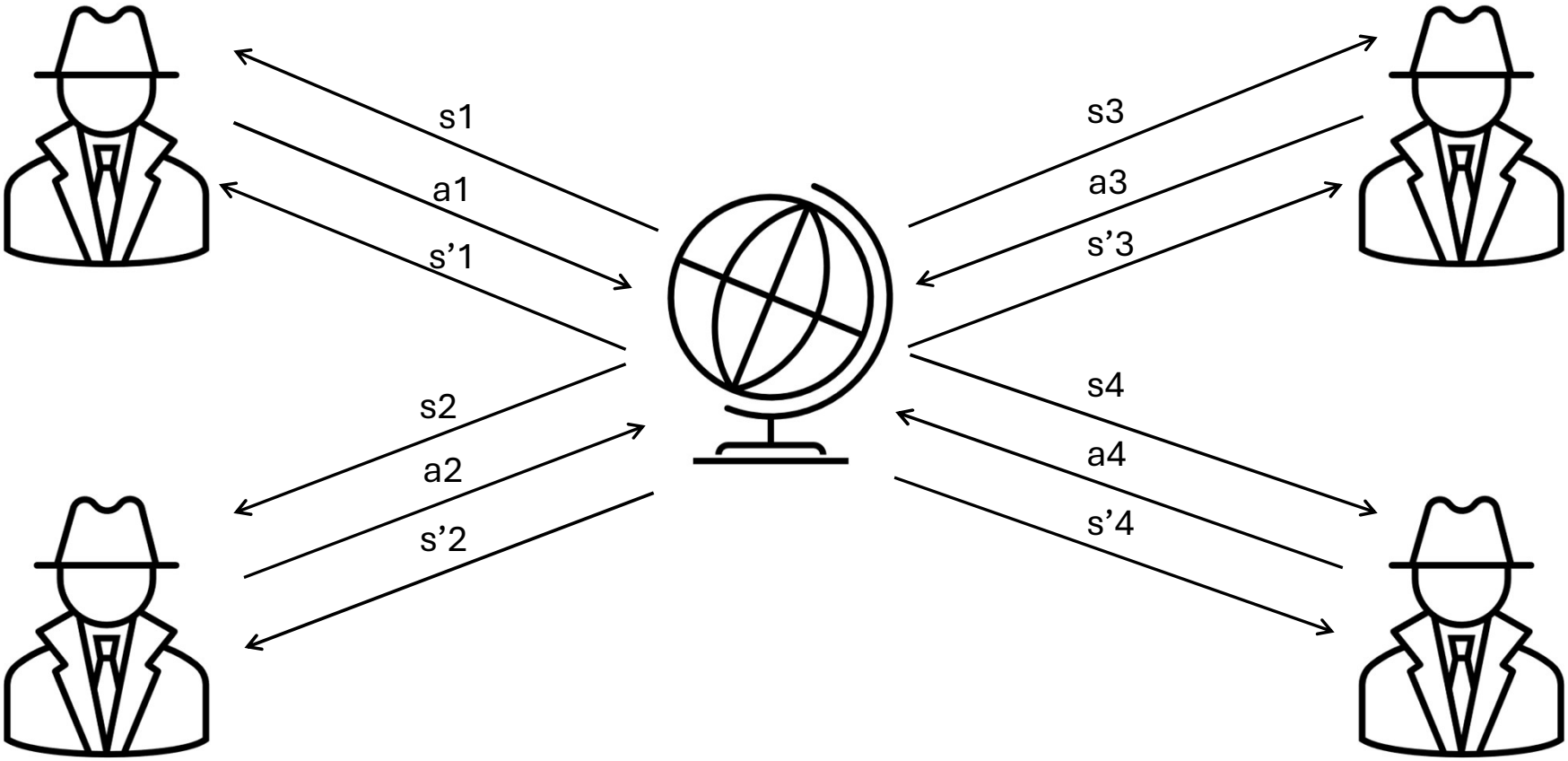

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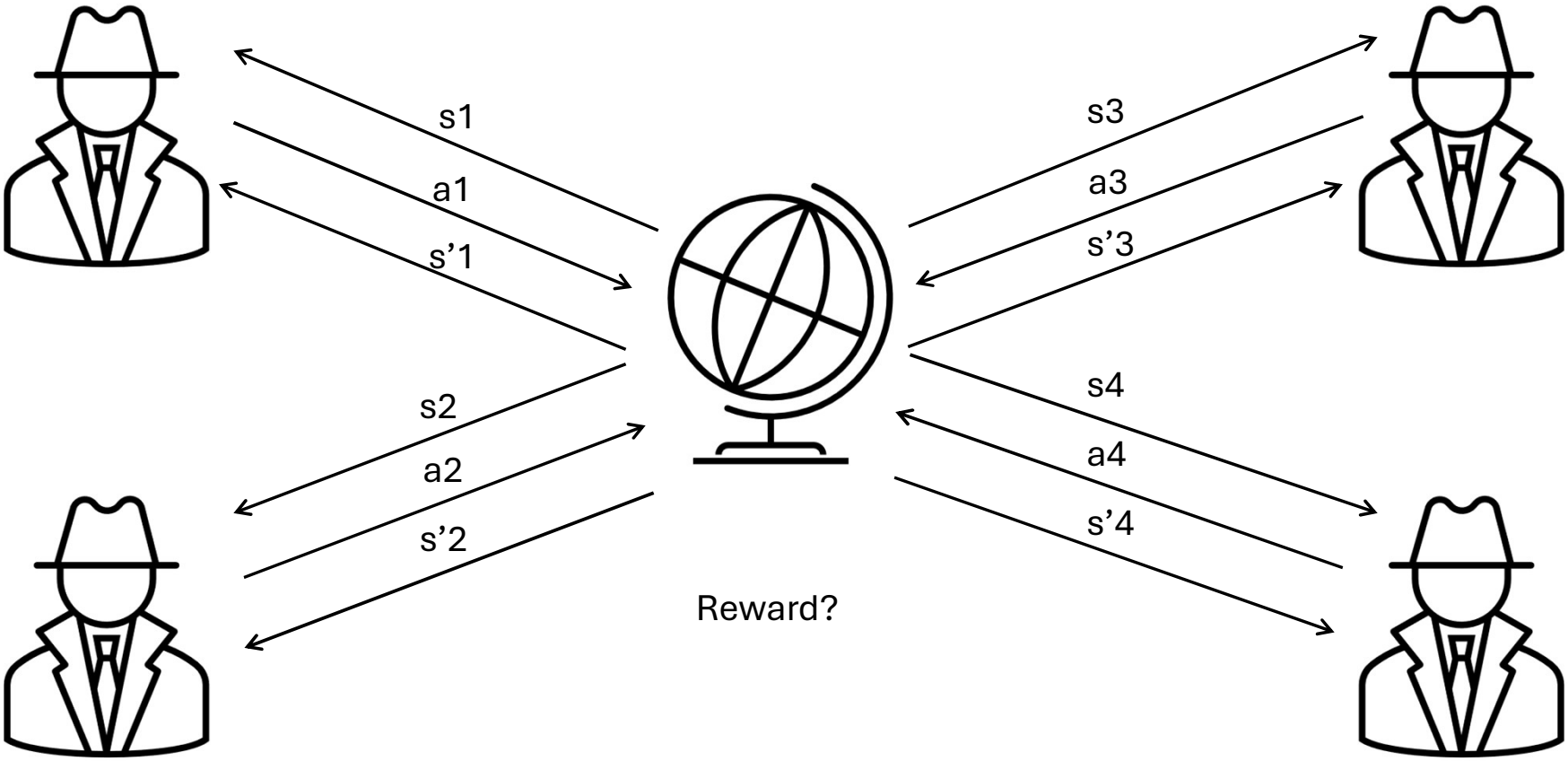


How to coordinate multiple agents?

Main challenge of the MARL is ***non-stationarity***.

Changes in the environment depend now not only on the action of the agent, but on the collective ***joint action***.

On top of that, it is not even obvious what do we want to achieve. Do we want to get the biggest rewards? Or maybe we want the rewards to be distributed equally among all agents?

Depending on the kind of reward signal we get and task at hand, we can coordinate the agents by the means

Incentive design

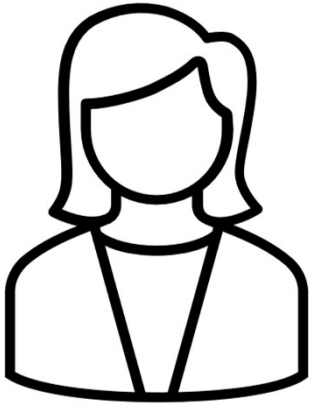
- Modifying the rewards in order to steer the joint policy
- Depending on the task at hand, we can incentivize cooperation, efficiency or the equality among agents
- It requires a central planner, or incentive designer whose purpose is to create reward function aligning agent objectives towards common goal

Independent agents

How we can approach the situation where agents make decisions independently, actions are costly and sometimes the best option would be just to… do nothing?

We can pose this problem as an instance of ***principal-agent model***, the most common framework in the contract theory.

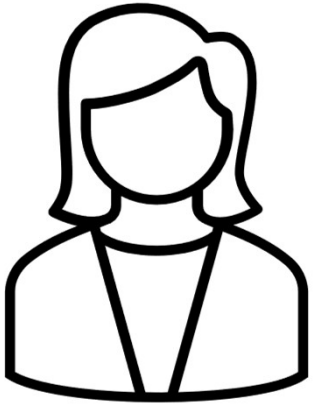
Contract Theory



Principal

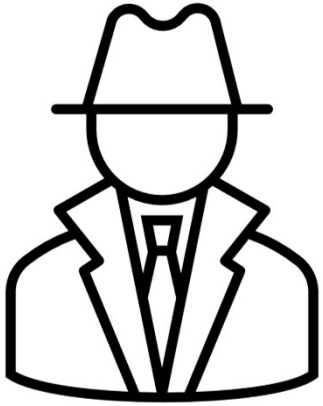
- Has a task to delegate
- Doesn’t possess the skillset to do it on her own
- Is willing to pay a share of the gain to have the task done by someone else

Contract Theory



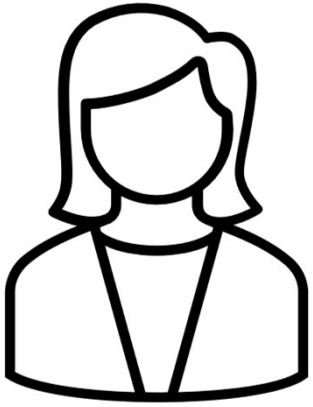
Principal

- Possess a skillset to do the task for principal
- Is willing to work if fairly compensated
- Otherwise will not complete the task, or will not put maximal efforts

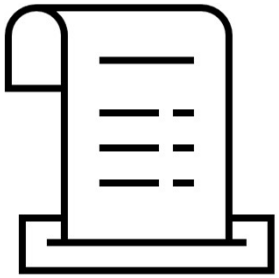


Agent

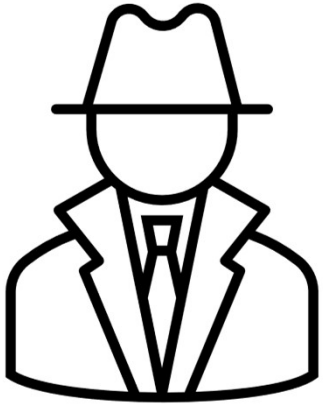
Contract Theory



Principal



To align their interests, they need some sort of a contract which outlines the rules of cooperation



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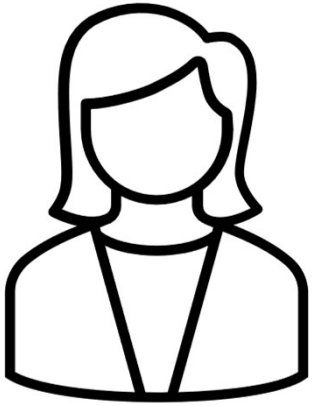
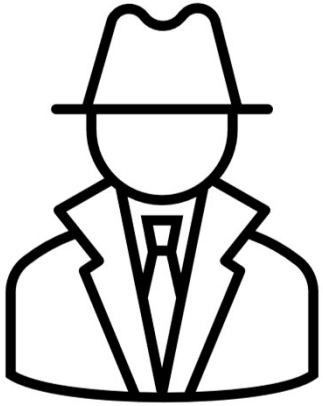
Contract Theory



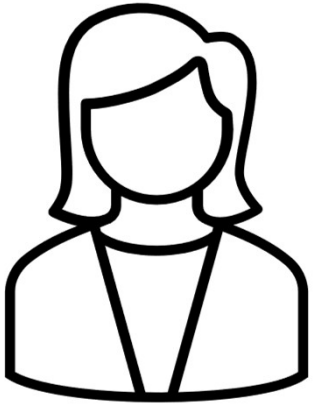
Principal

If an agent agrees to a contract, their cooperation begins



Agent

Contract Theory

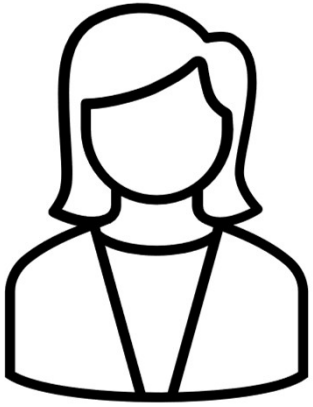


Principal

Agent

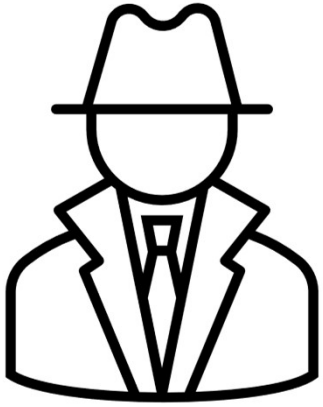
Agent makes (possibly costly) effort to complete a task

Contract Theory



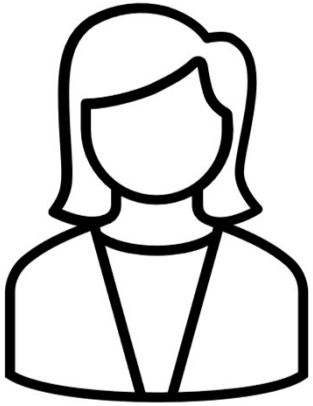
Principal

Agent transfers the produced gain to principal



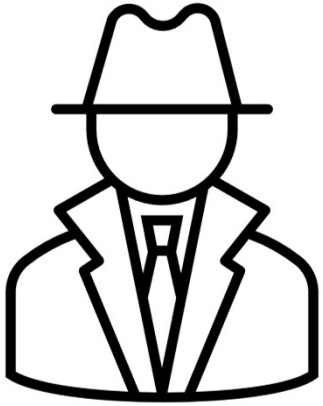
Agent

Contract Theory



Principal

Principal pays agent for the work



Agent

Real life examples

- Any work contract
- Insurance
- Public investments
- Professional sport
- Crowdsourcing
- Taxes

Basics of ACT

- Agent has actions $a_1, \ldots, a_n$, with underlying costs $0 \leq c_1 \leq \cdots \leq c_n$
- Agent's actions are unknown to the principal – she sees only outcomes
- Principal can collect reward from m outcomes $0 \leq r_1 \leq \cdots \leq r_m$
- Agent's actions lead to outcomes based on distribution F
- Principal offers agent a contract $\vec{t} = (t_1, \ldots, t_m)$, which conditions the payment on outcome
- Agent either accepts the contract and works for principal, or refuses it and plays do-nothing action a_0 , which has no cost

Constraints

- **Incentive compatibility** (IC) – agent's picks an action maximizing his utility
- **Individual rationality** (IR) – agent's expected utility is non-negative
- **Limited liability** (LL) – there are no negative transfers, i.e. contract payments are non-negative
- **Monotone likelihood ratio property** (MLRP) – actions with higher costs lead to outcomes with bigger rewards
- Any ties in agent's utilities are broken in favor of principal

Contracts in MARL

- We can think of a contract as a form of incentive design, which has additional layer of complexity – agents can agree to changes or not
- Central planner has to take it into account when designing contracts
- Another consideration to take into account, is whether the agents are hetero- or homogenous. Then the structure of contract might differ from agent to agent

Project Proposal: PARL

- Introduced by Ivanov et al. in the eponymous paper, is the first attempt to integrate contracting framework in reinforcement learning
- It leverages two level reinforcement learning, improving policy of principal and agent one at a time. Principal's action is a contract offered to agent before the actual move.

Algorithm 1 Meta-algorithm for finding SPE

```
Initialize the principal's policy ρ arbitrarily          ▷ e.g., ∀s ∈ S : ρ(s) = 0
while ρ not converged do                                  ▷ Inner-outer optimization loop
    Solve the agent's MDP: find π := π*(ρ)                ▷ Inner optimization level
    Solve the principal's MDP: find ρ := ρ*(π)            ▷ Outer optimization level
return (ρ, π)
```

Source: Ivanov et al., „Principal Agent Reinforcement Learning" https://arxiv.org/pdf/2407.18074v1

Project Proposal: PARL

Authors present both single- and multi-agent framework of learning the optimal contracts that aim to arrive at subgame perfect equilibrium (SPE).

What is SPE? It is a solution concept, which basically says, that the policies of two agents (players) in the sequential game form the Nash equilibrium in all subgames of the game – in other words, they best-respond to each other and there is no incentive to change strategy

Project Proposal - Implement

Your task for the project would be to implement the PARL for the single agent, and create an adequate MDP environment to test the algorithm against it.

In the second part of the project, you will implement the multi-agent variant of the algorithm to work with the Coin Game environment which I will provide.

Project Proposal - Extension

To extend the project, you will be able to choose from following:

- Limit the algorithm to the linear contract family (which represent the share of the potential reward)
- Test the algorithm on another environment compatible with PettingZoo
- Modify the objective – instead of focusing on SPE, try maximizing the welfare (sum of the rewards in the system) or fairness between the agents

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